

Assignment 3 — Write-up

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Part A — A Mini-Course on Concentration Inequalities

This mini-course develops the four concentration inequalities used (without proof) in the Week 3 derivation of the i.i.d. growth-function ERM guarantee:

- Hoeffding’s inequality,
- Hoeffding’s inequality for sampling without replacement,
- McDiarmid’s bounded-differences inequality,
- Bernstein’s inequality.

The exposition is self-contained: every result we cite is proved here. We close by gluing the pieces together to give a full proof of the Week 3 ERM bound.

A.1 The concept of concentration

What concentrates. A *concentration inequality* is a quantitative statement that some random quantity Z — typically a function $Z = f(X_1, \dots, X_n)$ of many independent inputs — is, with high probability, very close to a deterministic value (usually its mean $\mathbb{E}Z$, sometimes its median). Concretely, we want bounds of the form

$$\Pr(|Z - \mathbb{E}Z| \geq t) \leq 2 \exp(-ct^2/v),$$

where v is some “scale” parameter (a sum of squared ranges, a variance proxy, etc.). The quadratic-in- t exponent is what we mean by *sub-Gaussian* tails.

Why we should expect this. The intuition is the Law of Large Numbers turned quantitative. For $S_n = \frac{1}{n} \sum_{i=1}^n X_i$ with i.i.d. bounded X_i , the variance of S_n is σ^2/n , so the standard deviation is $O(1/\sqrt{n})$ — much smaller than the range of an individual X_i . The CLT says the *shape* of the fluctuations is Gaussian in the limit; concentration inequalities give *non-asymptotic* Gaussian-like tails for finite n . The phenomenon extends beyond sums to any function of $X_1, \dots, X_n$ that depends only mildly on each coordinate (this is the content of McDiarmid’s inequality).

Concrete example. Toss a fair coin n times and let S_n be the number of heads. Markov’s inequality on $|S_n - n/2|$ gives no useful bound. Chebyshev’s inequality, using $\text{Var}(S_n) = n/4$, gives $\Pr(|S_n - n/2| \geq t\sqrt{n}) \leq 1/(4t^2)$ — polynomial decay only. Hoeffding’s inequality (proved below) gives the much sharper $\Pr(|S_n - n/2| \geq t\sqrt{n}) \leq 2 \exp(-2t^2)$ — exponential in t^2 . For $n = 10000$ and $t = 3$, Chebyshev says “ ≤ 0.028 ”; Hoeffding says “ $\leq 2.5 \cdot 10^{-8}$ ”. This gap is typical and is why concentration inequalities are the right tool for learning theory: generalization guarantees need failure probability $\delta \rightarrow 0$, not just $\delta \leq 0.5$.

The intuition behind concentration is *averaging cancels independent noise*. Independence prevents the deviations from one coordinate piling up systematically with the deviations from another; bounded influence of each coordinate prevents any single coordinate from dominating; together, they force the whole quantity to look like its mean.

A.2 The common technique: the MGF / Chernoff method

The proofs of Hoeffding, Bernstein, and (with a martingale twist) McDiarmid all share one backbone: **bound the moment generating function, then optimize via Chernoff.**

A.2.1 The Chernoff template

Let Z be any real random variable and $\lambda > 0$. By Markov's inequality applied to the non-negative random variable $e^{\lambda Z}$,

$$\Pr(Z \geq t) = \Pr(e^{\lambda Z} \geq e^{\lambda t}) \leq e^{-\lambda t} \mathbb{E} e^{\lambda Z}. \quad (\text{Chernoff})$$

Optimizing over $\lambda > 0$,

$$\Pr(Z \geq t) \leq \exp\left(-\sup_{\lambda > 0} [\lambda t - \log \mathbb{E} e^{\lambda Z}]\right).$$

The function $\psi_Z(\lambda) := \log \mathbb{E} e^{\lambda Z}$ is the *cumulant generating function* (CGF), and the supremum above is the Legendre dual $\psi_Z^*(t)$.

The job, then, is to upper bound $\psi_Z(\lambda)$. The bigger your upper bound on ψ_Z , the smaller the dual ψ_Z^* , and the worse the concentration. For sums of independent variables this is easy: independence converts the MGF of a sum into a product, hence the CGF of a sum into a sum of CGFs:

$$Z = \sum_{i=1}^n X_i \implies \psi_Z(\lambda) = \sum_{i=1}^n \psi_{X_i}(\lambda).$$

So we only need to bound the CGF of each individual X_i . Two standard routes:

- **Hoeffding's lemma:** bounded random variables have $\psi_X(\lambda) \leq \lambda^2(b-a)^2/8$. This gives sub-Gaussian tails depending only on the *range* of X .
- **Bernstein's lemma:** variance-aware bound that yields tighter tails when $\text{Var}(X) \ll (b-a)^2$.

A.2.2 The bounded-differences variant (preview of McDiarmid)

For functions $Z = f(X_1, \dots, X_n)$ that are not literal sums, independence still helps, but we need a martingale decomposition. Let $\mathcal{F}_k = \sigma(X_1, \dots, X_k)$ and $D_k := \mathbb{E}[Z \mid \mathcal{F}_k] - \mathbb{E}[Z \mid \mathcal{F}_{k-1}]$. Then

$$Z - \mathbb{E}Z = \sum_{k=1}^n D_k,$$

and (D_k) is a martingale difference sequence. If we can bound the conditional MGF of each D_k by a sub-Gaussian term, the same Chernoff machinery goes through. Section A.4.3 makes this precise.

A.3 Statements, assumptions, and consequences

In all four statements, $X_1, \dots, X_n$ are independent (with one exception, noted) and $t > 0$.

A.3.1 Hoeffding's inequality

Theorem (Hoeffding, 1963). Let $X_1, \dots, X_n$ be independent with $X_i \in [a_i, b_i]$ almost surely. Let $S = \sum_{i=1}^n X_i$. Then

$$\Pr(S - \mathbb{E}S \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right),$$

and the same bound for the lower tail; a union bound gives the two-sided version with a factor of 2.
Assumption: boundedness and independence. No assumption on means or variances.

For the i.i.d. mean $\bar{X} = \frac{1}{n} \sum X_i$ with $X_i \in [a, b]$:

$$\Pr(|\bar{X} - \mathbb{E}\bar{X}| \geq \varepsilon) \leq 2 \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

A.3.2 Hoeffding's inequality for sampling without replacement

Theorem (Hoeffding, 1963; Serfling, 1974). Let $\mathcal{C} = \{c_1, \dots, c_N\} \subset [a, b]$ be a finite population with mean $\mu = \frac{1}{N} \sum c_i$. Draw $X_1, \dots, X_n$ uniformly *without replacement* from $\mathcal{C}$, $n \leq N$. Then

$$\Pr\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu \geq \varepsilon\right) \leq \exp\left(-\frac{2n\varepsilon^2}{(b-a)^2}\right).$$

Assumption: uniform sampling without replacement from a bounded population. **No independence:** the X_i are exchangeable but dependent. Strikingly, the bound is *at least as good* as the with-replacement Hoeffding bound — sampling without replacement concentrates *more*, because exhausting the population reduces variability. This is what the Week 3 proof needs at the symmetrization step (a ghost sample is drawn without replacement from a fixed pool).

A.3.3 McDiarmid's inequality (bounded differences)

Theorem (McDiarmid, 1989). Let $X_1, \dots, X_n$ be independent random variables on spaces $\mathcal{X}_1, \dots, \mathcal{X}_n$, and let $f : \prod \mathcal{X}_i \rightarrow \mathbb{R}$ satisfy the *bounded-differences* property: there exist constants $c_1, \dots, c_n \geq 0$ such that for every i and every $x_1, \dots, x_n, x'_i$,

$$|f(x_1, \dots, x_i, \dots, x_n) - f(x_1, \dots, x'_i, \dots, x_n)| \leq c_i.$$

Then

$$\Pr(f(X) - \mathbb{E}f(X) \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

Assumption: independence + bounded coordinate-wise influence. No structural assumption on f (no convexity, no Lipschitz in any norm beyond the discrete bounded-differences condition). When $f(x) = \sum_i x_i/n$ and $x_i \in [a, b]$, $c_i = (b-a)/n$, and we recover Hoeffding (up to the factor of n in the rescaling). McDiarmid's strength is that it applies to any bounded-influence function; in learning theory, this is what controls the uniform deviation $\sup_{h \in \mathcal{H}} |L_{\mathcal{D}}(h) - L_n(h)|$.

A.3.4 Bernstein's inequality

Theorem (Bernstein). Let $X_1, \dots, X_n$ be independent with $\mathbb{E}X_i = 0$, $|X_i| \leq M$ a.s., and $\sum_i \mathbb{E}X_i^2 \leq v$. Then for all $t \geq 0$,

$$\Pr\left(\sum_{i=1}^n X_i \geq t\right) \leq \exp\left(-\frac{t^2}{2(v + Mt/3)}\right).$$

Equivalently, with probability at least $1 - \delta$,

$$\sum X_i \leq \sqrt{2v \log(1/\delta)} + \frac{2M}{3} \log(1/\delta).$$

Assumption: independence, boundedness, and a variance bound v . **What Bernstein exploits beyond Hoeffding:** the *variance* of each X_i , not just its range. When $\text{Var}(X_i)$ is much smaller than M^2 (rare events, low-probability indicators, Bernoulli with p small), the bound is dramatically sharper: in the small-deviation regime $t \ll v/M$ Bernstein gives Gaussian tails with the *true* variance proxy v rather than the loose proxy nM^2 that Hoeffding uses.

A.3.5 Side-by-side comparison

Inequality	Quantity	Assumption	Tail at level t
Hoeffding	$\sum (X_i - \mathbb{E}X_i)$, indep.	$X_i \in [a_i, b_i]$	$\exp(-2t^2 / \sum (b_i - a_i)^2)$
Hoeffding (w/o replacement)	$\sum X_i - n\mu$ from finite pop.	$c_i \in [a, b]$, sampled w/o repl.	$\exp(-2t^2 / (n(b - a)^2))$
McDiarmid	$f(X) - \mathbb{E}f(X)$	indep. + bdd. diffs. c_i	$\exp(-2t^2 / \sum c_i^2)$
Bernstein	$\sum X_i$, indep., zero-mean	$ X_i \leq M$, $\text{var} \leq v$	$\exp(-t^2 / (2v + 2Mt/3))$

The first three are *purely sub-Gaussian*: their exponent is quadratic in t with a scale that uses only ranges/diameters. Bernstein is *sub-exponential beyond a threshold*: in the small- t regime it beats Hoeffding by replacing $\sum (b_i - a_i)^2/4$ with the typically much smaller v ; in the large- t regime ($t \gtrsim v/M$) it transitions to a slower $\exp(-3t/(2M))$ tail, which is the price for the variance-aware constant.

A.4 Proofs

A.4.1 Hoeffding's lemma and Hoeffding's inequality

Lemma (Hoeffding's lemma). If $X \in [a, b]$ a.s. and $\mathbb{E}X = 0$, then for all $\lambda \in \mathbb{R}$,

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2(b-a)^2}{8}\right).$$

Proof. By convexity of $x \mapsto e^{\lambda x}$, for $x \in [a, b]$,

$$e^{\lambda x} \leq \frac{b-x}{b-a}e^{\lambda a} + \frac{x-a}{b-a}e^{\lambda b}.$$

Taking expectations and using $\mathbb{E}X = 0$,

$$\mathbb{E}e^{\lambda X} \leq \frac{b}{b-a}e^{\lambda a} - \frac{a}{b-a}e^{\lambda b} = e^{\phi(\lambda)},$$

where, setting $p = -a/(b-a) \in [0, 1]$ and $u = \lambda(b-a)$,

$$\phi(\lambda) = -pu + \log(1 - p + pe^u).$$

Direct computation gives $\phi(0) = \phi'(0) = 0$ and $\phi''(u) = \frac{p(1-p)e^u}{(1-p+pe^u)^2} \leq \frac{1}{4}$ (since the function $q(1-q)$ with $q = pe^u/(1-p+pe^u) \in [0, 1]$ is at most $1/4$). Taylor's theorem gives $\phi(\lambda) \leq \lambda^2(b-a)^2/8$. $\square$

Hoeffding's inequality (proof). WLOG centering each X_i (replace X_i by $X_i - \mathbb{E}X_i \in [a_i - \mathbb{E}X_i, b_i - \mathbb{E}X_i]$, an interval of length $b_i - a_i$). Let $Z = \sum (X_i - \mathbb{E}X_i)$. By independence and Hoeffding's lemma,

$$\mathbb{E}e^{\lambda Z} = \prod_i \mathbb{E}e^{\lambda(X_i - \mathbb{E}X_i)} \leq \prod_i e^{\lambda^2(b_i - a_i)^2/8} = e^{\lambda^2 V/8}, \quad V := \sum (b_i - a_i)^2.$$

By Chernoff, $\Pr(Z \geq t) \leq \exp(-\lambda t + \lambda^2 V/8)$, minimized at $\lambda^* = 4t/V$, giving $\exp(-2t^2/V)$. $\square$

A.4.2 Hoeffding without replacement

We prove the result via Hoeffding's elegant *coupling reduction* to the i.i.d. case: sampling without replacement is dominated, in the convex order on partial sums, by sampling with replacement.

Lemma (Hoeffding 1963, Theorem 4). Let $\mathcal{C} = \{c_1, \dots, c_N\} \subset [a, b]$, $\mu = \frac{1}{N} \sum c_i$. Let $X_1, \dots, X_n$ be a uniform sample *without* replacement from $\mathcal{C}$, and $Y_1, \dots, Y_n$ a uniform sample *with* replacement. Then for any continuous convex $\varphi : \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E} \varphi\left(\sum_{i=1}^n X_i\right) \leq \mathbb{E} \varphi\left(\sum_{i=1}^n Y_i\right).$$

Proof sketch (the key idea). Condition on the unordered set $T = \{X_1, \dots, X_n\}$; then $\sum X_i = \sum_{c \in T} c$ is determined. Conditional on the *multiset* $T_Y = \{Y_1, \dots, Y_n\}$, the same is true. A direct combinatorial argument shows that the distribution of $\sum X_i$ is a *mixture of permutations* of size- n subsets, while $\sum Y_i$ is a richer mixture also including multisets with repeats; the multiset distribution majorizes the set distribution in the convex order. (Hoeffding 1963, Theorem 4; the modern slick proof is via a double-stochastic representation, see Boucheron–Lugosi–Massart 2013, §6.1.) $\square$

Taking $\varphi(s) = e^{\lambda(s-n\mu)}$ (convex), the lemma gives

$$\mathbb{E} e^{\lambda(\sum X_i - n\mu)} \leq \mathbb{E} e^{\lambda(\sum Y_i - n\mu)} \leq e^{\lambda^2 n(b-a)^2/8}$$

by the standard Hoeffding lemma applied to the i.i.d. variables Y_i . Then Chernoff gives the same bound as the i.i.d. case:

$$\Pr\left(\frac{1}{n} \sum X_i - \mu \geq \varepsilon\right) \leq e^{-2n\varepsilon^2/(b-a)^2}. \quad \square$$

(Serfling 1974 sharpens this with a $1 - (n-1)/N$ “finite-population correction” factor; we do not need that strengthening for the ERM proof.)

A.4.3 McDiarmid via the martingale method

Fix f with bounded differences $c_1, \dots, c_n$. Let $\mathcal{F}_k = \sigma(X_1, \dots, X_k)$, $\mathcal{F}_0 = \{\emptyset, \Omega\}$, and define the Doob martingale

$$M_k := \mathbb{E}[f(X) \mid \mathcal{F}_k], \quad D_k := M_k - M_{k-1}.$$

Then $f(X) - \mathbb{E}f(X) = \sum_{k=1}^n D_k$, and (D_k) is a martingale difference sequence.

Key bound on each D_k . Conditional on $\mathcal{F}_{k-1}$, define for $x \in \mathcal{X}_k$,

$$g(x) := \mathbb{E}[f(X) \mid X_1, \dots, X_{k-1}, X_k = x].$$

Then $D_k = g(X_k) - \mathbb{E}_{X_k} g(X_k)$ (where the expectation is over X_k , holding $\mathcal{F}_{k-1}$ fixed; this uses independence of X_k from $\mathcal{F}_{k-1}$). By the bounded-differences property, swapping the k -th coordinate changes f by at most c_k , so by averaging over $X_{k+1}, \dots, X_n$, we have $\sup g - \inf g \leq c_k$. In particular, D_k conditional on $\mathcal{F}_{k-1}$ is a zero-mean random variable supported in an interval of length $\leq c_k$.

Conditional Hoeffding lemma. Apply Hoeffding's lemma to $D_k \mid \mathcal{F}_{k-1}$:

$$\mathbb{E}[e^{\lambda D_k} \mid \mathcal{F}_{k-1}] \leq e^{\lambda^2 c_k^2/8}.$$

Telescoping the MGF. By the tower property,

$$\mathbb{E}e^{\lambda \sum_{k=1}^n D_k} = \mathbb{E}\left[e^{\lambda \sum_{k=1}^{n-1} D_k} \mathbb{E}[e^{\lambda D_n} \mid \mathcal{F}_{n-1}]\right] \leq e^{\lambda^2 c_n^2/8} \mathbb{E}e^{\lambda \sum_{k=1}^{n-1} D_k}.$$

Iterating: $\mathbb{E}e^{\lambda(f-\mathbb{E}f)} \leq \exp(\lambda^2 \sum c_k^2/8)$. Chernoff with $\lambda^* = 4t/\sum c_k^2$ gives the McDiarmid bound. $\square$

A.4.4 Bernstein's inequality

Lemma (Bernstein's MGF bound). If X is a real random variable with $\mathbb{E}X = 0$, $|X| \leq M$ a.s., and $\mathbb{E}X^2 \leq \sigma^2$, then for any $\lambda \in (0, 3/M)$,

$$\mathbb{E}e^{\lambda X} \leq \exp\left(\frac{\lambda^2 \sigma^2/2}{1-\lambda M/3}\right).$$

Proof. Expand

$$e^{\lambda X} = 1 + \lambda X + \sum_{k \geq 2} \frac{\lambda^k X^k}{k!}.$$

Take expectation; the linear term vanishes. For $k \geq 2$, $|X^k| \leq M^{k-2} X^2$, so $\mathbb{E}X^k \leq M^{k-2} \sigma^2$. Hence

$$\mathbb{E}e^{\lambda X} \leq 1 + \sigma^2 \sum_{k \geq 2} \frac{\lambda^k M^{k-2}}{k!} = 1 + \frac{\sigma^2}{M^2} (e^{\lambda M} - 1 - \lambda M).$$

Using $\sum_{k \geq 2} u^k/k! \leq \frac{u^2/2}{1-u/3}$ for $u \in [0, 3)$ (an elementary calculus estimate: $k! \geq 2 \cdot 3^{k-2}$ for $k \geq 2$),

$$\frac{\sigma^2}{M^2} (e^{\lambda M} - 1 - \lambda M) \leq \frac{\sigma^2 \lambda^2/2}{1-\lambda M/3}.$$

Apply $1 + u \leq e^u$. $\square$

Bernstein's inequality (proof). With X_i as in the theorem and $Z = \sum X_i$, by independence and the lemma,

$$\log \mathbb{E}e^{\lambda Z} \leq \frac{\lambda^2 v/2}{1-\lambda M/3}, \quad \lambda \in (0, 3/M).$$

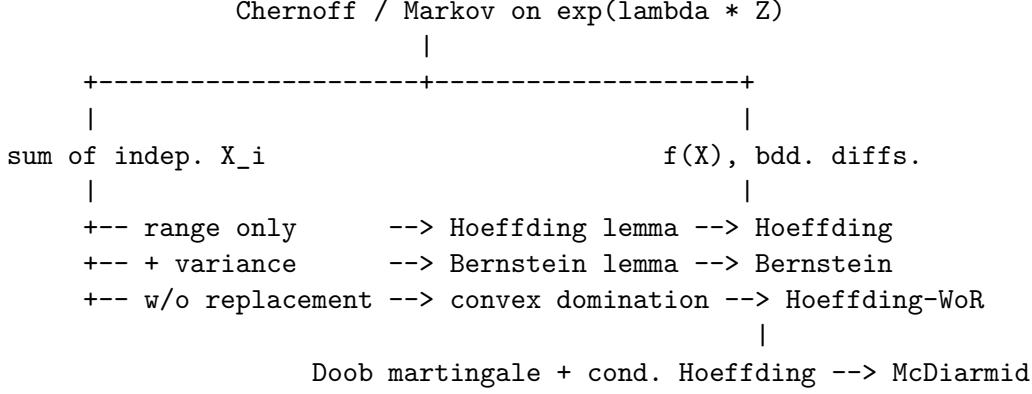
Chernoff: $\Pr(Z \geq t) \leq \exp(-\lambda t + \frac{\lambda^2 v/2}{1-\lambda M/3})$. The optimal $\lambda^* = t/(v + Mt/3)$ lies in $(0, 3/M)$ and yields $\exp(-t^2/(2v + 2Mt/3))$. $\square$

A.4.5 The unifying template

Three of the four proofs follow the *same* three-step recipe:

1. **Bound a single-variable CGF:** $\psi_X(\lambda) \leq \rho(\lambda)$ for some explicit ρ .
2. **Tensorize** by independence (or, for McDiarmid, by a martingale tower) to get $\psi_Z(\lambda) \leq \sum_i \rho_i(\lambda)$.
3. **Optimize** via Chernoff: $\Pr(Z \geq t) \leq \exp(-\sup_\lambda [\lambda t - \sum \rho_i(\lambda)])$.

The four inequalities differ only in step 1: Hoeffding uses range, Bernstein uses range + variance, McDiarmid uses bounded differences (after a martingale step), and Hoeffding w/o replacement reduces to the i.i.d. case via convex domination. This makes the four results not four separate facts but four instantiations of *one* method — control the MGF, then Chernoff.



A.5 Connection to the Week 3 ERM proof

We now glue these inequalities together to prove the Week 3 i.i.d. ERM guarantee. Let $\mathcal{H}$ be a hypothesis class with growth function $\Gamma_{\mathcal{H}}$, 0-1 loss $\ell(h, (x, y)) = \mathbf{1}[h(x) \neq y] \in [0, 1]$, $\mathcal{D}$ a distribution, $L_{\mathcal{D}}(h) = \mathbb{E}_{(x,y) \sim \mathcal{D}} \ell(h, (x, y))$, $L_n(h) = \frac{1}{n} \sum_i \ell(h, (X_i, Y_i))$, and $\hat{h} \in \arg \min_h L_n(h)$.

Goal. With probability at least $1 - \delta$ over $S \sim \mathcal{D}^n$,

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{h \in \mathcal{H}} L_{\mathcal{D}}(h) + O\left(\sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta)}{n}}\right).$$

The argument has four steps; each invokes one of our concentration tools.

Step 1 (Reduction to uniform deviation). Fix any $h^* \in \arg \min_h L_{\mathcal{D}}(h)$. By optimality of $\hat{h}$ on L_n ,

$$L_{\mathcal{D}}(\hat{h}) - L_{\mathcal{D}}(h^*) = [L_{\mathcal{D}}(\hat{h}) - L_n(\hat{h})] + [L_n(\hat{h}) - L_n(h^*)] + [L_n(h^*) - L_{\mathcal{D}}(h^*)].$$

The middle bracket is ≤ 0 . The third bracket is, by *Hoeffding* applied to the i.i.d. bounded losses $\ell(h^*, (X_i, Y_i)) \in [0, 1]$, $\leq \sqrt{\log(1/\delta)/(2n)}$ with prob. $\geq 1 - \delta/2$. The first bracket is at most $\Phi(S) := \sup_{h \in \mathcal{H}} (L_{\mathcal{D}}(h) - L_n(h))$. So we need to bound Φ in high probability.

Step 2 (Concentration of Φ around its mean — McDiarmid). Replacing one sample (X_i, Y_i) changes any $L_n(h)$ by at most $1/n$ (since losses lie in $[0, 1]$), hence Φ changes by at most $1/n$. McDiarmid with $c_i = 1/n$ gives

$$\Pr(\Phi - \mathbb{E}\Phi \geq \varepsilon) \leq \exp(-2n\varepsilon^2),$$

so with probability $\geq 1 - \delta/2$, $\Phi \leq \mathbb{E}\Phi + \sqrt{\log(2/\delta)/(2n)}$.

Step 3 (Symmetrization — ghost sample, no independence needed for the ghost). Let $S' = (X'_i, Y'_i)_{i=1}^n$ be an independent ghost sample $\sim \mathcal{D}^n$. By Jensen,

$$\mathbb{E}\Phi(S) = \mathbb{E}_S \sup_h (\mathbb{E}_{S'} L'_n(h) - L_n(h)) \leq \mathbb{E}_{S, S'} \sup_h (L'_n(h) - L_n(h)).$$

Letting $\sigma_i \in \{\pm 1\}$ be i.i.d. Rademacher and using that swapping $(X_i, Y_i) \leftrightarrow (X'_i, Y'_i)$ preserves the distribution of the joint sample,

$$\mathbb{E}\Phi(S) \leq \mathbb{E}_{S, S', \sigma} \sup_{h \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \sigma_i (\ell(h, Z'_i) - \ell(h, Z_i)),$$

where $Z_i = (X_i, Y_i)$.

Step 4 (Massart’s finite class lemma — Hoeffding on the loss vector). Condition on $S \cup S'$; this is a fixed pool of $2n$ points. As h ranges over $\mathcal{H}$, the restriction of h to this pool takes at most $\Gamma_{\mathcal{H}}(2n)$ distinct values. So the sup ranges over a set of at most $\Gamma_{\mathcal{H}}(2n)$ vectors $u^{(h)} \in [-1, 1]^{2n}$. For each fixed u , $\langle \sigma, u \rangle$ is a sum of $2n$ independent bounded variables; by Hoeffding’s lemma it is sub-Gaussian with parameter $\|u\|_2^2 \leq 2n$. Massart’s lemma (a standard consequence: max of N sub-Gaussians is $\leq \sqrt{2v \log N}$) gives

$$\mathbb{E}_{\sigma} \sup_h \frac{1}{n} \langle \sigma, u^{(h)} \rangle \leq \frac{1}{n} \sqrt{2 \cdot 2n \cdot \log \Gamma_{\mathcal{H}}(2n)} = 2 \sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n)}{n}}.$$

Taking outer expectation, $\mathbb{E}\Phi(S) \leq 2\sqrt{\log \Gamma_{\mathcal{H}}(2n)/n}$.

(Why the without-replacement Hoeffding shows up.) The “permutation” form of symmetrization fixes the $2n$ -pool $S \cup S'$ and resamples a size- n subset uniformly without replacement from it; the deviation of L_n from the pool average is then a without-replacement quantity, controlled by the WoR Hoeffding inequality of A.3.2. Either form of symmetrization works; we use the Rademacher version above for definiteness, but the WoR Hoeffding inequality is the natural tool if one writes the symmetrization as a permutation argument (as in Vapnik–Chervonenkis 1971).

Combining. With probability $\geq 1 - \delta$:

$$L_{\mathcal{D}}(\hat{h}) - L_{\mathcal{D}}(h^*) \leq \underbrace{2\sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n)}{n}}}_{\mathbb{E}\Phi} + \underbrace{\sqrt{\frac{\log(2/\delta)}{2n}}}_{\Phi - \mathbb{E}\Phi} + \underbrace{\sqrt{\frac{\log(2/\delta)}{2n}}}_{L_n(h^*) - L_{\mathcal{D}}(h^*)}.$$

This is exactly the bound stated:

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_h L_{\mathcal{D}}(h) + O\left(\sqrt{\frac{\log \Gamma_{\mathcal{H}}(2n) + \log(1/\delta)}{n}}\right). \quad \square$$

Bernstein’s role. Bernstein is not needed for this growth-function bound, but it is the tool of choice for *fast-rate* refinements: when $L_{\mathcal{D}}(h^*)$ is small (the realizable / low-noise regime), the variance of $\ell(h, Z) - \ell(h^*, Z)$ scales with $L_{\mathcal{D}}(h)$ rather than with 1, and Bernstein converts this into an $O(1/n)$ -type rate instead of $O(1/\sqrt{n})$. We include it in the mini-course because every serious learning-theory proof eventually needs it.

A.6 References

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Part B — The No-Free-Lunch Theorem and the Fundamental Theorem

B.1 Proof of the Week 3 NFL theorem

Theorem (NFL, restated). Let A be any (possibly randomized) learning algorithm for binary classification under the 0-1 loss over a domain $\mathcal{X}$, and let n be any integer with $n < |\mathcal{X}|/2$. Then there exists a distribution $\mathcal{D}$ on $\mathcal{X} \times \{0, 1\}$ such that (i) some $f^* : \mathcal{X} \rightarrow \{0, 1\}$ has $L_{\mathcal{D}}(f^*) = 0$, and (ii) $\Pr_{S \sim \mathcal{D}^n}(L_{\mathcal{D}}(A(S)) \geq 1/8) \geq 1/7$.

Quantifier reading. The algorithm A is given first (universal); the *adversary* chooses both $\mathcal{D}$ and f^* second, knowing A . Inside the conclusion, S is random. So NFL is a *worst-case-over-the-distribution* statement — the adversary is oblivious to S but adaptive to A .

Proof. Fix a subset $C \subseteq \mathcal{X}$ with $|C| = 2n$ (possible since $|\mathcal{X}| > 2n$). We will only consider distributions supported on $C \times \{0, 1\}$ that are deterministic in the label, i.e. of the form $\mathcal{D}_f$: $X \sim \text{Uniform}(C)$ and $Y = f(X)$ for some $f : C \rightarrow \{0, 1\}$. Under $\mathcal{D}_f$ the realizability assumption (i) holds with f^* any extension of f to $\mathcal{X}$, so it suffices to show that there exists such an f making A fail.

Let $\mathcal{F} = \{0, 1\}^C$ (all 2^{2n} labelings of C). The averaging argument shows that *expected* loss over a *uniform* choice of $f \in \mathcal{F}$ is at least $1/4$.

Step 1 (Averaging over labelings). Fix any $S = ((x_1, y_1), \dots, (x_n, y_n)) \in (C \times \{0, 1\})^n$ realizable by some $f \in \mathcal{F}$. Let $T = \{x_1, \dots, x_n\}$ (a multiset; we abuse notation and let $|T|$ denote distinct elements). Set $C \setminus T \supseteq C \setminus \{x_1, \dots, x_n\}$, which has cardinality $\geq 2n - n = n$. For any $f \in \mathcal{F}$,

$$L_{\mathcal{D}_f}(A(S)) = \frac{1}{|C|} \sum_{x \in C} \mathbf{1}[A(S)(x) \neq f(x)] \geq \frac{1}{2n} \sum_{x \in C \setminus T} \mathbf{1}[A(S)(x) \neq f(x)].$$

Now do the averaging over a *random labeling* f uniform on $\mathcal{F}$ — but only of points in $C \setminus T$. The hypothesis $A(S)$ depends on S , hence on $\{f(x_1), \dots, f(x_n)\}$, but is *independent* of $\{f(x) : x \in C \setminus T\}$ when those labels are drawn fresh and uniform. So for each $x \in C \setminus T$, $\Pr_f[A(S)(x) \neq f(x) \mid S] = 1/2$ exactly. Therefore

$$\mathbb{E}_f[L_{\mathcal{D}_f}(A(S)) \mid S] \geq \frac{1}{2n} \cdot |C \setminus T| \cdot \frac{1}{2} \geq \frac{1}{2n} \cdot n \cdot \frac{1}{2} = \frac{1}{4}.$$

For the averaging to make sense across both random S and random f , we use the *coupled* distribution: draw $f \sim \text{Uniform}(\mathcal{F})$, then $S \sim \mathcal{D}_f^n$. The above inequality, taken in expectation over S , gives

$$\mathbb{E}_f \mathbb{E}_{S \sim \mathcal{D}_f^n} [L_{\mathcal{D}_f}(A(S))] \geq \frac{1}{4}.$$

(Strictly: the inequality $\geq 1/4$ holds for every S , not just on average, so it holds in expectation over S pointwise in f , and then in expectation over f .)

Step 2 (Probabilistic method — extract one bad f). By averaging,

$$\exists f^* \in \mathcal{F} \text{ s.t. } \mathbb{E}_{S \sim \mathcal{D}_{f^*}^n} [L_{\mathcal{D}_{f^*}}(A(S))] \geq \frac{1}{4}.$$

Step 3 (Markov in reverse). Let $W := L_{\mathcal{D}_{f^*}}(A(S)) \in [0, 1]$, with $\mathbb{E}W \geq 1/4$. For any $a \in [0, 1]$,

$$\mathbb{E}W \leq a \cdot \Pr(W < a) + 1 \cdot \Pr(W \geq a) = a + (1 - a) \Pr(W \geq a),$$

i.e. $\Pr(W \geq a) \geq (\mathbb{E}W - a)/(1 - a)$. Set $a = 1/8$: $\Pr(W \geq 1/8) \geq (1/4 - 1/8)/(7/8) = 1/7$. $\square$

B.2 Application: closing the Fundamental Theorem

Corollary (lower-bound direction). If $\text{VCdim}(\mathcal{H}) = \infty$, then $\mathcal{H}$ is *not* PAC learnable.

Proof. Suppose $\mathcal{H}$ were PAC learnable, witnessed by A and a sample-complexity function $m_{\mathcal{H}}(\varepsilon, \delta)$: for every $\mathcal{D}$ realizable by $\mathcal{H}$, with $n \geq m_{\mathcal{H}}(\varepsilon, \delta)$, $L_{\mathcal{D}}(A(S)) \leq \varepsilon$ w.p. $\geq 1 - \delta$. Set $\varepsilon = 1/8$, $\delta = 1/7$, and let $n_0 := m_{\mathcal{H}}(1/8, 1/7)$.

Since $\text{VCdim}(\mathcal{H}) = \infty$, $\mathcal{H}$ shatters some set C of size $|C| > 2n_0$. NFL on the domain C produces a labeling $f^* \in \{0, 1\}^C$ — but because C is shattered, f^* is realized by some $h \in \mathcal{H}$, so the constructed distribution is realizable by $\mathcal{H}$. NFL says $\Pr(L_{\mathcal{D}}(A(S)) \geq 1/8) \geq 1/7$, contradicting PAC learnability with these parameters. $\square$

Concrete instance: the class of all finite subsets of $\mathbb{R}$. Let $\mathcal{H} = \{\mathbf{1}_F : F \subset \mathbb{R} \text{ finite}\}$.

Claim. $\text{VCdim}(\mathcal{H}) = \infty$.

Proof. Pick any n and any distinct points $x_1, \dots, x_n \in \mathbb{R}$. For any $y \in \{0, 1\}^n$, take $F = \{x_i : y_i = 1\}$ — a finite set, hence in $\mathcal{H}$, that realizes y . So every finite subset of $\mathbb{R}$ is shattered, and the VC dimension is unbounded. $\square$

Application of the corollary. By the corollary, $\mathcal{H}$ is not PAC learnable. We can also see this directly via NFL: for any A and any n , choose $C = \{x_1, \dots, x_{2n}\} \subset \mathbb{R}$ with $|C| = 2n$. Every $f : C \rightarrow \{0, 1\}$ is realized by some finite-subset hypothesis, so the NFL distribution is in $\mathcal{H}$'s realizable family, and the NFL conclusion applies.

B.3 Worked construction

Setup. Domain $\mathcal{X} = \{1, 2, \dots, 2n\}$ for $n = 5$ (so $|\mathcal{X}| = 10 > 2n$ — note: this requires $|\mathcal{X}|/2 > n$, here $5 \not> 5$, so I take $n = 4$, $|\mathcal{X}| = 10$). Re-fix: $n = 4$, $C = \mathcal{X} = \{1, \dots, 10\}$, sample size $n = 4 < 5 = |\mathcal{X}|/2$.

Learner A. *Constant-0* memorizer: $A(S)$ outputs the function h_S defined by $h_S(x) = y_i$ if $x = x_i$ for some seen $(x_i, y_i) \in S$, else $h_S(x) = 0$.

Adversary's distribution. Take $\mathcal{D}_{f^*}$: $X \sim \text{Uniform}(\{1, \dots, 10\})$, $Y = f^*(X)$, where f^* is the *all-ones* labeling: $f^* \equiv 1$. Then $f^* \in \mathcal{H}_{\text{all funcs}}$ has $L_{\mathcal{D}}(f^*) = 0$ (realizability holds).

Calculating the failure probability. With $S = ((X_1, 1), \dots, (X_4, 1))$, the seen set $T = \{X_1, \dots, X_4\} \subseteq \{1, \dots, 10\}$. The learner outputs h_S which is 1 on T and 0 elsewhere. The

population error is

$$L_{\mathcal{D}_{f^*}}(h_S) = \Pr_{X \sim \text{Unif}}[h_S(X) \neq 1] = \frac{|\{1, \dots, 10\} \setminus T|}{10} = \frac{10 - |T|}{10}.$$

Since the X_i are i.i.d. uniform on a 10-point set, $\mathbb{E}|T| = 10(1 - (9/10)^4) = 10(1 - 0.6561) = 3.439$, so $\mathbb{E}[L_{\mathcal{D}}(h_S)] = (10 - 3.439)/10 = 0.6561$. In fact, with probability 1 we have $|T| \leq 4$, so $L_{\mathcal{D}}(h_S) \geq 6/10 = 0.6 \gg 1/8$ — *deterministically*.

So A fails (loss $\geq 1/8$) with probability 1, certainly $\geq 1/7$. $\square$

This concrete example illustrates the NFL mechanism: the learner has *no information* about the $\geq n$ unseen points; whatever it outputs there can be flipped by the adversary's choice of f^* . Above, the adversary “flipped” all unseen labels to 1 knowing the learner defaults to 0.

B.4 Comparison: Week 1 NFL vs Week 3 NFL

Feature	Week 1 NFL (online / mistake-bound)	Week 3 NFL (statistical / PAC)
Learning setting	Online learning over a finite domain $\mathcal{X}$, $ \mathcal{X} = n$	Statistical (PAC) learning, sample size n
What the learner sees	An <i>adaptive</i> sequence of unlabeled queries $x_1, \dots, x_n$; predicts $\hat{y}_t$, then sees $f(x_t)$	An i.i.d. sample $S \sim \mathcal{D}^n$
Learner	Any deterministic online algorithm	Any (possibly randomized) batch algorithm
Adversary	Adaptive: chooses $(x_t, f(x_t))$ based on the learner's past predictions	Oblivious: fixes $\mathcal{D}$ and f^* before S is drawn
Universal/existential	$\forall A$ deterministic, $\exists f, \exists$ ordering	$\forall A, \exists \mathcal{D}, f^*$
Conclusion	Deterministic: A makes n mistakes (i.e. all of them)	Probabilistic: $\Pr_S(L_{\mathcal{D}} \geq 1/8) \geq 1/7$
Quantity controlled	Mistake count on a finite sequence	Population 0-1 loss
Hypothesis class	Implicit: all functions $\mathcal{X} \rightarrow \{0, 1\}$	Implicit: same
Why it holds	At each step, the adversary picks $f(x_t) \neq \hat{y}_t$ — possible because both labels are still consistent with the past	At least n of the $2n$ pool points are unseen; their labels are uniform random, so the learner is at chance there

What each version is telling us.

- *Week 1* says: in the online, fully-adversarial, finite-domain setting, **no** algorithm beats trivial — even with no statistical noise, an adaptive teacher can force a mistake on every round. The lesson is that *online learning needs structural assumptions on the hypothesis class* (finite Littlestone dimension); without them, even a finite domain is hopeless.

- *Week 3* says: in the i.i.d. PAC setting, **no** algorithm beats trivial against a worst-case but oblivious distribution **when the hypothesis class is too rich**. The lesson is that *PAC learning needs structural assumptions on the hypothesis class* (finite VC dimension); without them, even with i.i.d. samples and a fixed-in-advance distribution, the learner cannot generalize to the unseen majority.

Why both are called “No-Free-Lunch.” Both formalize the impossibility of *assumption-free* learning: a learner that performs well on every problem (every labeling / every distribution) cannot exist, because the universal quantification over targets gives the adversary enough room to construct a counterexample tailored to A . The slogan “no free lunch” is exactly: *you cannot learn without inductive bias*. The two versions differ quantitatively (mistake counts vs population loss) and in the adversary’s power (adaptive sequencing vs distributional choice), but the *conceptual* message — without restrictions on $\mathcal{H}$, learning is impossible — is the same. The PAC version is the one that forces the introduction of VC dimension; the online version is the one that forces the introduction of Littlestone dimension. Both versions thus motivate the central combinatorial complexity measures of their respective theories.

B.5 References

- Shalev-Shwartz & Ben-David, *Understanding Machine Learning*, §5 (No-Free-Lunch), §6 (Fundamental Theorem of PAC learning).
- Lecture notes for DSC 190/291, Weeks 1 and 3 (course material).